



main

burn / crates / burn-import / SUPPORTED-ONNX-OPS.md



Msa360 Feat: Add Cosh and Sinh (#2959)

dae36cc · 2 weeks ago



393 lines (389 loc) · 27.5 KB

Preview

Code

Blame

Raw



Supported ONNX Operators

Note: some ONNX Ops listed below are pseudo Ops, such as Linear , Conv1d , Conv2d (or other with 1d, 2d suffixes used signify the dimensionality). These are not real ONNX Ops, but are used to represent the corresponding Burn Op.

ONNX OP	Import Support	Burn Support
Abs	✓	✓
Acos	✗	✗
Acosh	✗	✗
Add	✓	✓
And	✗	✗
ArgMax	✓	✓
ArgMin	✗	✗
Asin	✗	✗
Asinh	✗	✗
Atan	✗	✗
Atanh	✗	✗
AveragePool1d	✓	✓
AveragePool2d	✓	✓

ONNX OP	Import Support	Burn Support
BatchNormalization	✓	✓
Bernoulli	✗	✗
BitShift	✗	✗
BitwiseAnd	✗	✗
BitwiseNot	✗	✗
BitwiseOr	✗	✗
BitwiseXor	✗	✗
BlackmanWindow	✗	✗
Cast	✓	✓
CastLike	✗	✗
Ceil	✗	✗
Celu	✗	✗
CenterCropPad	✗	✗
Clip	✓	✓
Col2Im	✗	✗
Compress	✗	✗
Concat	✓	✓
ConcatFromSequence	✗	✗
Constant	✓	✓
ConstantOfShape	✓	✓
Conv1d	✓	✓
Conv2d	✓	✓
Conv3d	✓	✓
ConvInteger	✗	✗
ConvTranspose1d	✓	✓
ConvTranspose2d	✓	✓
ConvTranspose3d	✓	✓

ONNX OP	Import Support	Burn Support
Cos	✓	✓
Cosh	✓	✓
CumSum	✗	✗
DepthToSpace	✗	✗
DequantizeLinear	✗	✗
Det	✗	✗
DFT	✗	✗
Div	✓	✓
Dropout	✓	✓
DynamicQuantizeLinear	✗	✗
Einsum	✗	✗
Elu	✗	✗
Equal	✓	✓
Erf	✓	✓
Exp	✓	✓
Expand	✓	✓
EyeLike	✗	✗
Flatten	✓	✓
Floor	✓	✓
Gather	✓	✓
GatherElements	✓	✓
GatherND	✗	✗
Gelu	✓	✓
Gemm	✓	🟡
GlobalAveragePool	✓	✓
GlobalLpPool	✗	✗
GlobalMaxPool	✗	✗

ONNX OP	Import Support	Burn Support
Greater	✓	✓
GreaterOrEqual	✓	✓
GridSample	✗	✗
GroupNormalization	✗	✓
GRU	✗	✓
HammingWindow	✗	✗
HannWindow	✗	✗
Hardmax	✗	✗
HardSigmoid	✓	✓
HardSwish	✗	✗
Identity	✓	✓
If	✗	✓
Im	✗	✗
InstanceNormalization	✗	✓
IsInf	✗	✗
IsNaN	✗	✗
LayerNormalization	✓	✓
LeakyRelu	✓	✓
Less	✓	✓
LessOrEqual	✓	✓
Linear	✓	✓
Log	✓	✓
LogSoftmax	✓	✓
Loop	✗	✗
LpNormalization	✗	✗
LpPool	✗	✗
LRN	✗	✗

ONNX OP	Import Support	Burn Support
LSTM	✗	✓
MatMul	✓	✓
MatMulInteger	✗	✓
Max	✓	✓
MaxPool1d	✓	✓
MaxPool2d	✓	✓
MaxRoiPool	✗	✗
MaxUnpool	✗	✗
Mean	✓	✓
MeanVarianceNormalization	✗	✗
MelWeightMatrix	✗	✗
Min	✓	✓
Mish	✗	✗
Mod	✗	✗
Mul	✓	✓
Multinomial	✗	✗
Neg	✓	✓
NegativeLogLikelihoodLoss	✗	✗
NonMaxSuppression	✗	✗
NonZero	✗	✗
Not	✓	✓
OneHot	✓	✓
Optional	✗	✗
OptionalGetElement	✗	✗
OptionalHasElement	✗	✗
Or	✗	✗
Pad	✓	✓

ONNX OP	Import Support	Burn Support
Pow	✓	✓
PRelu	✓	✓
QLinearConv	✗	✗
QLinearMatMul	✗	✗
QuantizeLinear	✗	✗
RandomNormal	✓	✓
RandomNormalLike	✓	✓
RandomUniform	✓	✓
RandomUniformLike	✓	✓
Range	✓	✓
Reciprocal	✓	✓
ReduceL	✗	✗
ReduceLogSum	✗	✗
ReduceLogSumExp	✗	✗
ReduceMax	✓	✓
ReduceMean	✓	✓
ReduceMin	✓	✓
ReduceProd	✓	✓
ReduceSum	✓	✓
ReduceSumSquare	✗	✗
Relu	✓	✓
Reshape	✓	✓
Resize	✓	✓
ReverseSequence	✗	✗
RNN	✗	✓
RoiAlign	✗	✗
Round	✗	✗

ONNX OP	Import Support	Burn Support
Scan	✗	✗
Scatter	✗	✓
ScatterElements	✗	✗
ScatterND	✗	✗
Selu	✗	✗
SequenceAt	✗	✗
SequenceConstruct	✗	✗
SequenceEmpty	✗	✗
SequenceErase	✗	✗
SequenceInsert	✗	✗
SequenceLength	✗	✗
SequenceMap	✗	✗
Shape	✓	✓
Shrink	✗	✗
Sigmoid	✓	✓
Sign	✓	✓
Sin	✓	✓
Sinh	✓	✓
Size	✗	✗
Slice	✓	✓
Softmax	✓	✓
SoftmaxCrossEntropyLoss	✗	✗
Softplus	✗	✗
Softsign	✗	✗
SpaceToDepth	✗	✗
Split	✓	✓
SplitToSequence	✗	✗

ONNX OP	Import Support	Burn Support
Sqrt	✓	✓
Squeeze	✓	✓
STFT	✗	✗
StringNormalizer	✗	✗
Sub	✓	✓
Sum	✓	✓
Tan	✓	✓
Tanh	✓	✓
TfidfVectorizer	✗	✗
ThresholdedRelu	✗	✗
Tile	✓	✓
TopK	✓	✓
Transpose	✓	✓
Trilu	✓	✓
Unique	✗	✗
Upsample	✗	✗
Where	✓	✓
Xor	✗	✗
Unsqueeze	✓	✓

 Supported via other operations.